

MOHAMED AAFRITH

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SUMMARY

AI Full Stack Developer / Software Engineer with **1.5+ years of experience** designing, developing, and deploying scalable **LLM and AI powered applications**. Strong expertise in **Python, FastAPI, Flask, Django, React, REST APIs**, and **cloud-native architectures**. Experienced in building **LLM-driven reasoning and automation workflows, retrieval-augmented pipelines, computer vision, NLP**, and **LLM fine-tuning**, with hands-on experience integrating AI agents into enterprise systems and deploying models in production environments on **AWS** and **GCP**.

EXPERIENCE

Temporary Academic Instructor - University of Sri Jayewardenepura

Dept. of Electrical and Electronic Engineering, Faculty of Engineering, Sri Lanka

July 2026 – Present

- Deliver undergraduate lectures, lab sessions, and tutorials to Electrical and Electronic Engineering students, covering programming, software engineering, and applied computing topics.
- Conduct assignments, tutorials, and practical exercises, and evaluate student performance through assessments, lab work, and project supervision.

AI Full Stack Developer / Software Engineer - SCOPULE FZC

UAE(remote)

January 2025 – February 2026

- Designed, developed, and deployed multiple full-stack web applications including e-commerce platforms, POS systems, and custom client solutions, handling end-to-end frontend, backend, and database architecture using React.js, Node.js / Python (FastAPI), RESTful APIs, and SQL/NoSQL databases.
- Designed and integrated LLM-driven reasoning and tool-calling workflows using the Gemini API, enabling multi-step task automation, structured content generation, and intelligent decision-making within production applications. Fine-tuned LLaMA 3.2 on domain-specific datasets to ground agent responses in business context, achieving 90%+ accuracy.
- Integrated advanced computer vision solutions into web applications using Hugging Face, OpenCV, and MediaPipe, and built robust execution logic (input validation, error handling, and fallback flows) to ensure reliable end-to-end task completion.
- Built a full-stack, automation-driven stall reservation system for a large-scale book fair using Flask and React, orchestrating a multi-step workflow spanning JWT-based authentication, real-time stall availability, QR code generation, and automated email notifications, along with admin dashboards for analytics, revenue tracking, and reservation management.

Software Engineer (Internship) - SLT Mobitel Pvt Ltd

Colombo, Sri Lanka

August 2024 – February 2025

- Developed a full-stack seat reservation system using Flask and PostgreSQL, serving 100+ internal users for real-time seat booking and admin role management, integrating secure authentication via Google OAuth and designing REST APIs for downstream service integration.
- Containerized and deployed the Flask application using Docker and Google Cloud Run, building CI/CD pipelines for continuous deployment and managing database migrations with SQLAlchemy, Flask-Migrate, and Alembic.
- Built an API-driven automation pipeline integrating with the Bitbucket REST API to retrieve, aggregate, and visualize repository analytics on a live dashboard, reducing manual reporting time by 60%.

PROJECTS

Emotion and Speech Aware Human Computer Interaction System

2025

- Created a real-time emotion-driven HCI system using Python, OpenCV, and MediaPipe, integrating facial expression recognition, hand-gesture mouse control, and voice command processing.
- Added ML pipelines for emotion classification, real-time inference, and multimodal user interaction through a unified desktop interface. Achieved 88.5% emotion classification accuracy and deployed the system as a standalone executable with real-time performance.

Learning System with Eye and Emotion Tracking

2025

- Built a full-stack learning platform with role-based access, live video sessions, and real-time interactions using Agora and WebSockets, supporting interactive classes, attendance tracking, and multi-user collaboration.

- Integrated gaze tracking and on-device emotion inference to generate engagement metrics, alerts, and analytics reports via a FastAPI + MongoDB backend. Implemented real-time dashboards to visualize student engagement, session highlights, and AI-driven recommendations for instructors to improve learning outcomes.

Smart Money Management System

2025

- Engineered a secure full-stack expense management system using FastAPI + MongoDB Atlas and React (Vite) + Tailwind, featuring JWT authentication and role-based access.
- Implemented expense tracking, advanced filtering, and real-time analytics dashboards with smart input pipelines for SMS, receipt scanning, and voice-based expense capture using AI-assisted extraction.
- Trained a YOLO-based OCR model on a custom bill image dataset for automatic bill uploading and detail extraction. Fine-tuned a LLaMA 3.2 model on bank and notification messages, achieving 95% accuracy, and integrated it with a speech-to-text pipeline for voice-based expense entries.

CERTIFICATIONS

Certificate in Machine Learning Specialization [\[Link\]](#)

Certificate in Deep Learning Specialization [\[Link\]](#)

AWS Cloud Technical Essentials [\[Link\]](#)

ACHIEVEMENTS

Top 10, IEEE Final Year Project Competition

Dec 2025

IEEE, University of Ruhuna

2nd Runners-up, Haxtreme 2.0 Hackathon

Nov 2023

IEEE Student Branch, Sri Lanka

Paper Accepted, ICARC 2026

2026

06th International Conference on Advanced Research in Computing

TECHNICAL SKILLS

Programming Languages : Python, JavaScript, TypeScript

Agentic AI / LLM : LLM Fine-tuning (LLaMA), Tool/Function Calling, Prompt Engineering, RAG, Gemini API, Hugging Face, Structured Outputs

Frontend Development : React, ReactJS, HTML, CSS, JavaScript, TailwindCSS

Backend Development : REST APIs, Microservices, API Security, Scalable Systems, Django, FastAPI, Flask

AI / ML : Machine Learning, Deep Learning, NLP, Computer Vision, YOLO, OCR

Databases : MySQL, PostgreSQL, MongoDB

Cloud & DevOps : Docker, Google Cloud Platform, AWS, CI/CD

Version Control : Git, GitHub, Bitbucket

EDUCATION

Faculty Of Engineering, University of Ruhuna, Galle, Sri Lanka

BSc (Hons) in Engineering – Computer Engineering / 2021 – 2025

- CGPA: 3.21/4.0 (Second Lower Division)
- Relevant Coursework: Machine Learning, Neural Networks, Data Structures, Software Architecture, Blockchain and Cyber Security

REFERENCES

References available upon request